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!pip install -q opencv-python matplotlib numpy scikit-image
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import cv2
import numpy as np
import matplotlib.pyplot as plt
from skimage import exposure
from skimage.io import imread

# function to display images
def show_side_by_side(title, img1, img2, lab1="Method 1", lab2="Method 2"):
    plt.figure(figsize=(10,4))

    plt.subplot(1,2,1)
    plt.imshow(img1, cmap='gray')
    plt.title(lab1)
    plt.axis("off")

    plt.subplot(1,2,2)
    plt.imshow(img2, cmap='gray')
    plt.title(lab2)
    plt.axis("off")

    plt.suptitle(title, fontsize=14)
    plt.show()
```

```
# Paths to degraded images
image_paths = [
    "/content/Picture1.png",
    "/content/Picture2.png",
    "/content/Picture3.png",
    "/content/Picture4.png",
    "/content/Picture5.png"
]

# Load reference image for histogram matching
reference_img = cv2.imread(image_paths[0], 0) # use Picture1 as reference

print("Using Picture1 as histogram reference.\n")

for path in image_paths:
    print("Processing:", path)

    img = cv2.imread(path, 0)
    if img is None:
        print("Error: File not found")
        continue

    # ----- 1) Histogram Equalization -----
    heq = cv2.equalizeHist(img)

    # ----- 2) Histogram Matching -----
    # skimage expects float image
    img_float = img.astype(np.float32) / 255.0
    ref_float = reference_img.astype(np.float32) / 255.0
    hmatch = exposure.match_histograms(img_float, ref_float)
    hmatch = np.uint8(hmatch * 255)

    # ----- SHOW RESULTS -----
    show_side_by_side(
        title=f"Histogram Enhancement - {path}",
        img1=heq,
        img2=hmatch,
        lab1="Histogram Equalization",
        lab2="Histogram Matching"
    )

    print("\nDone for:", path, "\n\n")
```

